

5kVA RESIDENTIAL SOLAR SYSTEM

Consumer Safety & 2026 Compliance Data Sheet

Standard: ISO 20607 | Compliance: Zimbabwe Energy Governance Framework 2026

1. 2026 REGULATORY CONFORMANCE

This system is engineered to meet the **Energy Solar Products and Installation Regulations 2026**.

- **Performance Standard:** PV modules meet the Minimum Energy Performance Standard (MEPS) for 2026 ($> 20.5\%$ efficiency).
- **Labelling:** Mandatory 2026 Star-Rating Labeling utilized for transparent consumer comparison.
- **Licensing:** Installed by ZERA-certified technicians under the 2026 Installer Competence Mandate.

2. SYSTEM ARCHITECTURE & SAFETY LIMITS

NatfortBlue		
Inverter	5kVA Hybrid (Pure Sine Wave)	IEC 62109-1 / 62109-2
Battery Bank	48V LiFePO4 (Lithium)	UN38.3 (Thermal Safety)
PV Array	450W - 550W Tier 1 Panels	IEC 61215 (Environmental)
Protection	AC/DC Surge Protectors (SPD)	SI 2026 Surge Mandate

3. CRITICAL SAFETY PROCEDURES (EMERGENCY)

A. Emergency Rapid Shutdown (The "60-Second" Rule)

In case of fire, flood, or electrical smell, follow this sequence immediately:

1. **PV ISOLATOR:** Turn the DC Breaker from panels to **OFF**.
2. **BATTERY MCB:** Flip the heavy-duty Battery Breaker to **OFF**.
3. **AC INPUT:** Switch off the ZESA/Grid Input breaker.
4. **AC OUTPUT:** Switch off the Main House Feed breaker.

B. Visual Indicator Warnings

- **Solid Red Light (ALARM):** Internal fault. Shutdown system and call support.
- **Flashing Blue/Green:** Normal operation / Charging.
- **DANGER:** "Sweet" or Chemical Odor indicates potential Lithium thermal event. **Evacuate room.** Do not use water. Use Class D or Dry Powder only.

Yamhiro (Shona)

Kudzima Nechimuchimbidze: 1. PV Isolator (OFF). 2. Battery MCB (OFF). 3. AC Input (OFF). 4. AC Output (OFF). *Kana ukanzwa kunhuwirira kwemakemikari, buda mumba nekukasira!*

Iziyabizo (Ndebele)

Ukucitsha Ngokushesha: 1. PV Isolator (OFF). 2. Battery MCB (OFF). 3. AC Input (OFF). 4. AC Output (OFF). *Uma unuka iphunga lama-chemical, phuma endlini masinyane!*

4. 2026 CONSUMER PROTECTION RIGHTS

- **Verification of Origin:** Access to the manufacturer's certificate of conformity.
- **Efficiency Guarantee:** Minimum 10-year linear performance warranty.
- **End-of-Life Plan:** Mandatory Lithium Recycling Path provided by Natfort Energy.

5. MAINTENANCE & PERFORMANCE

- **Dust Management:** Clean panels every 30 days during dry seasons (Harare/Bulawayo).
- **Load Management:** Avoid exceeding 4,500W. Use high-surge items sequentially.

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CHIKAMU CHECHENGETEDZO NEKUTEVERA MUTEMO

Kusimbisa ruzivo kumutengi wese pasi peFramework ye2026

3. NZIRA DZEKUTEVERA MUNGOZI (EMERGENCY)

Mutengi wese anofanira kunzwisisa mashandisirwo emagetsi zvizere.

A. Kudzima Gadziriro Nechimuchimbidze (The "60-Second" Rule)

Kana paita moto, mafashamo emvura, kana kunhuwa kwezvemagetsi, tevera matanho aya:

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PV ISOLATOR: Dzima breaker rezuva (rinobva kumapanel) riende pa **OFF**.

BATTERY MCB: Dzima breaker hombe rebhatiri riende pa **OFF**.

AC INPUT: Dzima breaker remagetsi anobva kuZESA kana mugridhi.

AC OUTPUT: Dzima breaker guru rinoendesa magetsi mumba.

B. Zviratidzo Zveyambiro (Visual Indicators)

Chiedza Chitsvuku (ALARM): Pane dambudziko. Dzima zvese ufonere **Natfort Support**.

Chiedza Chebhuruu/Chitemamwenje: Gadziriro iri kushanda zvakanaka.

Kunhuwa kweMakemikari (NGOZI): Bhatiri rinogona kunge riri kupisa (*Thermal event*). **Budai mumba mose!** Musashandise mvura. Shandisai Class D extinguisher chete.

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INGXENYE YEZOKUPHEPHA LOKULANDELA UMTHEHO

Ukuqinisa ulwazi kumthengi wonke ngaphansi kwe-Framework ye-2026

3. INDLELA ZOKUZIVIKELA ENGOZINI (EMERGENCY)

Wonke umthengi kumele azwisise ngezokuphepha ngolimi lwakhe.

A. Ukucitsha Isistimu Ngokushesha (The "60-Second" Rule)

Uma kulo mlilo, izikhukhula, loba unuka iphunga likagesi, landela lezi zinyathelo:

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PV ISOLATOR: Citsha ibreaker yelanga (evela kumapanel) iye ku-**OFF**.

BATTERY MCB: Citsha ibreaker elikhulu lebhatri liye ku-**OFF**.

AC INPUT: Citsha ibreaker yomlilo ovela koZESA (grid input).

AC OUTPUT: Citsha ibreaker elikhulu eliyisa amagetsi endlini.

B. Izibane Zeziyabizo (Visual Indicators)

Isibane Esibomvu (ALARM): Kulenkinga phakathi kwasistimu. Citsha konke ubusufonela **Natfort Support** masinyane.

Isibane Esiluhlaza/Esikhanyayo: Isistimu isebenza kahle loba iyachaja.

Iphunga lama-Chemical (INGOZI): Kungaba libhatri le-Lithium elitshisa kakhulu. **Phumani endlini lonke!** Lingasebenzisi amanzi. Sebenzisani i-Class D loba i-Dry Powder extinguisher kuphela.

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